TOPOSCOUT

SITE INTELLIGENCE REPORT

GNSS SURVEY PLANNING SYSTEM · SITE INTELLIGENCE REPORT

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES	ELEVATION	STATE PLANE	FEMA ZONE	DECLINATION
38.3459°N 122.3047°W WGS84	**66 ft** NAVD88 · MSL	**CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83)** —	**Zone X** 06055C0510FF	**13.0° E** WMM 2026

BEST GNSS WINDOW **Sun, Jul 12 · 2:00 PM – 6:00 PM** | **PDOP 0.97 · 35 sv** | **NEAREST CORS** **P261 · 14.1 mi ✓**

REPORT CONTENTS

SCAN FOR KMZ
Site file · Google Earth
↓ Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
—

ASSESSOR
Napa County Assessor · 1127 First St, Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

SITE & SIGNAL REPORT

COORDINATES	ELEVATION	MAGNETIC DECLINATION	FEMA FLOOD ZONE
38.3459, -122.3047	**66 ft**	**13.0° E**	**X**
WGS84 / NAD83	Above MSL (approx)	Compass reads 13.0° too far W	AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX
4.7

AVG KP
2.6

SOLAR FLUX F10.7
138 sfu

TEC (VTEC) ⚠ DEGRADED
19.4 TECU
-4.7 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.6). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Sun, Jul 12	0.97	2:00 PM – 6:00 PM	35	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location — orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates. For site-specific weather, terrain mask angle, and nearest CORS station, order an On-Demand Site Report.

🔥 WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 12, 2026) · Conditions may change — verify before fieldwork

🛰 GPS CONSTELLATION HEALTH

⚠ 1 Satellite Currently Unhealthy — PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

🌿 Land Cover & Multipath Risk

SKY OBSTRUCTION

MULTIPATH RISK

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.9–0.9 across survey window

PDOP remains ≤ 0.9 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

Sunday, Jul 12

WEEKEND

Overcast | 💨 10 mph | GO

High 90°F | Low 56°F | Wind max 10 mph | Gusts 12 mph | Precip 0.00" | Sunrise 5:56 AM | Sunset 8:33 PM

🌘 Waning Crescent 6%

GNSS: Avg PDOP **0.97** · Best 4-hr window 2:00 PM – 6:00 PM (PDOP 0.93, 35 sats avg)

GNSS / DOP — HOURLY

	PDOP	SATS	RATING
6:00 AM	0.87	36 sv	IDEAL
7:00 AM	0.91	34 sv	IDEAL
8:00 AM	1.06	29 sv	GOOD
9:00 AM	1.00	33 sv	GOOD
10:00 AM	1.00	29 sv	GOOD
11:00 AM	1.06	27 sv	GOOD
12:00 PM	0.96	29 sv	IDEAL
1:00 PM	1.00	28 sv	GOOD
2:00 PM	0.90	34 sv	IDEAL
3:00 PM	0.98	33 sv	IDEAL
4:00 PM	0.98	34 sv	IDEAL
5:00 PM	0.86	38 sv	IDEAL
6:00 PM	1.01	32 sv	GOOD
7:00 PM	1.04	30 sv	GOOD

⛅ **GO — GOOD CONDITIONS**

Partly cloudy | Wind: 10.1 mph (gusts 11.9 mph) | Precip: 0" | Temp: 56.3°–90°F

🚁 DRONE WIND ANALYSIS

✅ **GO** Peak: 10.1/11.9 mph S

☀ Sunrise: **5:56 AM** ☾ Sunset: **8:33 PM** fly window clipped to daylight

✅ Best flight window: **6:00 AM – 9:00 PM (15 hrs)**

🔄 Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph ⚡=turbulence	DIR
5 AM	3/5 mph	SSE
6 AM	4/7 mph	ESE
7 AM	2/3 mph	ESE
8 AM	3/3 mph	W
9 AM	3/4 mph	SW
10 AM	5/5 mph	SSW
11 AM	6/7 mph	SSE
12 PM	5/6 mph	S
1 PM	6/6 mph	S
2 PM	7/7 mph	S
3 PM	7/7 mph	S
4 PM	10/11 mph	SSW
5 PM	9/11 mph	SSW
6 PM	9/11 mph	SSW
7 PM	8/11 mph	S
8 PM	7/12 mph	SSE

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS A** ▶ FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
✈ TFR STATUS ✅ **NO TFRs WITHIN 30 MI** Check B4UFLY ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

AI FIELD ANALYSIS — CLAUDE HAIKU 4.5

FIELD PLANNING REPORT

Napa County, CA | 38.3459, -122.3047

Survey Date: Sunday, July 12, 2026

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VERDICT: GO

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Primary rationale: Excellent PDOP geometry (avg 0.97, best window 0.93 with 35 satellites), stable moderate ionospheric activity manageable with standard static protocols, and favorable weather conditions (no precipitation, light winds). Unhealthy PRN 11 already excluded from constellation calculations and does not materially degrade geometry.

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BEST SURVEY WINDOW

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Time: 2:00 PM – 6:00 PM (local time, Sunday July 12)

Rationale: Peak satellite geometry (PDOP 0.93, 35 satellites from GPS+GLONASS+Galileo+BeiDou constellation). Window occurs during afternoon when atmospheric conditions stabilize post-morning boundary layer rise. Provides 4-hour continuous operational window with minimal hour-to-hour PDOP variance.

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BEST 4-HOUR STATIC GNSS SESSION

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Recommended: 2:00 PM – 6:00 PM (or any 4-hour continuous subset within this window)

Session parameters:

— Elevation mask: 13° (site-specific terrain recommendation)

— Occupation target: 120 minutes baseline (moderate ionospheric activity; extend to 150 min if sub-cm accuracy required)

— Expected horizontal precision: ±10 mm + 1 ppm baseline length (static mode, moderate TEC)

— Expected vertical precision: ±15 mm + 1 ppm baseline length

— Constellation: 35 satellites (PDOP 0.93 maintains strong redundancy despite PRN 11 outage)

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DRONE OPERATIONS ASSESSMENT

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Status: APPROVED with conditions.

Wind: Maximum 10 mph — well within operational limits for most survey-grade UAS platforms. No constraint.

Weather: Overcast — no precipitation, visibility adequate for visual line-of-sight (VLOS) operations.

Recommendation: Conduct aerial reconnaissance or photogrammetry during the same 2:00 PM – 6:00 PM window to coincide with ground control point establishment. Overcast conditions may reduce contrast in orthophoto output; plan accordingly.

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FIELD SAFETY WARNINGS

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— No active wildfires detected within reporting range (all-clear).

— Air quality excellent (AQI 45, "Good"); no respiratory or visibility hazard.

— Temperature range 56–90°F: adequate hydration and sun protection recommended for 4-hour static session; no heat stress concern.

— Terrain: Dwarf scrub cover; low multipath risk and low sky obstruction. Standard site setup protocols sufficient.

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SUMMARY

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Sunday, July 12, 2026 is a strong operational day for GNSS survey and drone support. Geometry is excellent (PDOP <1.0 all day, peak 0.93 in afternoon), ionospheric delay is below normal (TEC 19.4 TECU) and manageable with standard occupation times, and weather is stable with no precipitation and light winds. The recommended survey window 2:00 PM – 6:00 PM provides optimal satellite geometry and atmospheric conditions. PRN 11 is unhealthy but has been auto-excluded from constellation calculations and does not affect day ranking. Proceed with fieldwork as planned; extend static sessions to 150 minutes if centimeter-level vertical accuracy is required.

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk
1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor
1127 First St, Napa, CA 94559

⚡ Nearby Control Points · 38.3459, -122.3047

CORS STATIONS — WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
NGS ↗	**P261**	HUNTERHILLCN2004	14.1 mi
NGS ↗	**P198**	PETALUMAIRCN2004	17.5 mi
NGS ↗	**OHLN**	OHLONE PARK	23.5 mi
NGS ↗	**P196**	MEACHUMLFLCN2006	24.0 mi
NGS ↗	**CASR**	SANTA ROSA CA	24.9 mi
NGS ↗	**P267**	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS — WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗	**JT9010**	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9018**	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9009**	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9017**	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9034**	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
Datasheet ↗	**JT9012**	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
Datasheet ↗	**JT9043**	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗					

FEMA FLOOD ZONE

X — AREA OF MINIMAL FLOOD HAZARD

Panel: **06055C0510FF**
Effective: Sep 29, 2010
View FEMA Map ↗

DATA COLLECTOR IMPORT — NGS BENCHMARKS

Copy All

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────────────────────────────────────────────────────────────────────────
NGS BENCHMARK CONTROL POINTS — Copy / Paste to Data Collector
Datum: NAD83  |  Vertical: NAVD88  |  Source: NGS NDE API
────────────────────────────────────────────────────────────────────────
PID      NAME                    LAT           LON           ELEV_FT  ORDER
────────────────────────────────────────────────────────────────────────
JT9010   13                        38.348333    -122.302778       62.7
JT9018   13 E                      38.348333    -122.302778       59.0
JT9009   13                        38.346667    -122.301111       58.7
JT9017   13 D                      38.345000    -122.301111       58.7
JT9034   20 A                      38.345000    -122.309444       68.0
JT9012   13                        38.351667    -122.304444       62.5
JT9043   25 A                      38.340000    -122.302778       61.4
JT9016   13 D                      38.341667    -122.297778       59.7
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🌊 SITE CONDITIONS · STREAM GAUGES & WATER LEVELS

USGS STREAM GAUGES — within 25 miles

STATION	DIST	LEVEL	FLOW	FLOOD STATUS
NAPA R NR NAPA CA #11458000	1.5 mi	2.36 ft	4.08 cfs	● Normal
NATHANSON C A SONOMA CA #11458600	9.3 mi	—	—	● Normal
SONOMA C A AGUA CALIENTE CA #11458500	10.4 mi	8.84 ft	2.73 cfs	● Normal
NAPA R NR ST HELENA CA #11456000	14.1 mi	1.44 ft	3.43 cfs	● Normal
SONOMA CREEK A KENWOOD CA #11458433	14.9 mi	3.93 ft	0.19 cfs	● Normal
PUTAH C NR WINTERS CA #11454000	16.8 mi	7.85 ft	578 cfs	— Unknown
SUISUN SLOUGH NR BENICIA CA #11455740	19.6 mi	14.40 ft	-4,650 cfs	— Unknown
POPE C A WALTER SPRINGS CA #11453590	21.1 mi	—	—	— Unknown

USGS LAKES & RESERVOIRS — within 150 miles

STATION	DIST	LEVEL	FULL POOL	LINK
CLEAR LK A LAKEPORT CA #11450000	58.1 mi	5.05 ft	N/R	USGS ↗

CALIFORNIA RESERVOIRS (DWR/CDEC) — within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	697.01 ft (-9 ft)	706 ft	36K af	**68%**
Folsom Lake	66.5 mi	448.64 ft (-17 ft)	466 ft	792K af	**81%**
Lake Oroville	93.4 mi	862.11 ft (-38 ft)	900 ft	2,871K af	**81%**
New Melones Lake	100.1 mi	1,020.25 ft (-68 ft)	1,088 ft	1,664K af	**69%**
Black Butte Lake	101.2 mi	—	228 ft	74K af	**55%**

USBR RESERVOIRS (Bureau of Reclamation) — within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP

ARMY CORPS RESERVOIRS (USACE) — within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	697 ft	695.01 ft	—	**100%**
Camanche	89 mi	230.3 ft	221.86 ft	—	**104%**
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	**101%**
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	**107%**
Black Butte-Pool	101 mi	457 ft	458.36 ft	—	**100%**
Don Pedro	137 mi	—	N/R	—	**—**

USGS GROUNDWATER WELLS — within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL

004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

⬡ AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Coast Ranges Sedimentary aquifer (N100CACSTL) consists of fractured sedimentary rocks formed during the Tertiary period in a marine depositional setting. This formation underlies the Napa Valley floor and foothills, with water stored in fractures and interstices of sandstone, siltstone, and shale layers. At this site, water availability depends on fracture connectivity and seasonal recharge from local precipitation and lateral flow, with moderate to variable yields typical of consolidated sedimentary aquifers.

FORMATION

Name	*Coast Ranges Sedimentary (Tertiary)*
Age / Rock	*Tertiary age; fractured sedimentary rocks including sandstone, siltstone, and shale*
Extent	*2,500–3,500 sq mi (est. USGS/DWR)*

DEPTH & RECHARGE

Depth Range	*51–295 ft (5–30 floors)*
Typical Depth	*80–200 ft (8–20 floors)*
Recharge Source	*Precipitation on upland outcrops and hillslopes; lateral groundwater flow from higher-elevation recharge areas*
Transit Time	*20–50 years to 100 ft depth (est. USGS/DWR)*

WATER QUALITY

pH	*est. 6.8–7.5 (est. USGS/DWR)*
Hardness	*est. 150–350 mg/L $CaCO_3$ (est. USGS/DWR)*
TDS	*est. 250–550 mg/L (est. USGS/DWR)*
Notes	*No sample data — typical for formation. Expect variable quality due to fractured sedimentary matrix; local geochemistry may reflect marine paleodeposits.*

TREND

Trend	*Declining water levels from 1977 to 2014 (9.90–87.71 ft BGS range), with partial recovery by 2022 (21.86 ft BGS at one well). Multi-decadal downward trend suggests sustained regional groundwater demand exceeding recharge.*
Source	*USGS NWIS and DWR well-monitoring data; 1977 baseline and 2014–2022 recent measurements*
Seasonal Swing	*est. 10–25 ft (est. USGS/DWR)*

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

⬡ AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The Castle Rock-Napa Valley area (aquifer code N100CACSTL) overlies consolidated bedrock and alluvial deposits with well-drained soils, characteristics that provide natural resistance to compaction-induced subsidence. This site is situated outside any State Water Resources Control Board-designated critically overdrafted groundwater basin, indicating that historical groundwater extraction has not created the excessive drawdown necessary to trigger clay layer consolidation. Surveyors working at this location can rely on stable vertical datums without subsidence correction factors.

SUBSIDENCE ASSESSMENT

Susceptibility	*Low*
Primary Mechanism	*None - consolidated terrain*
Annual Rate	*<0.5 mm/yr (est. USGS)*
Historic Loss	*None documented*
Aquifer Compaction	*None - aquifer not over-pumped*
SGMA Status	*Not applicable - not in SGMA basin*

BENCHMARK IMPLICATIONS

Monument Risk	*Low - monuments stable*
Elevation Validity	*Good - no active subsidence*
GNSS Recommendation	*Standard benchmark verification adequate*

SURVEY ACTION

Required Action	*No special action required*
Dataset	*CA DWR SGMA Bulletin 118 2024.1*
Coverage	*California critically overdrafted basins only - full US coverage pending*

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

Site Class	D (default)
Risk Category	II

WIND / SNOW / ICE / FROST (ASCE 7-22)

Basic Wind Speed	92 mph
Ground Snow Load	5.8 psf
Winter Wind (W2)	0.35
Ice Thickness	None
Frost Depth	**12 in minimum**

Est. from IECC Zone 3C — AFI data unavailable
Verify with local building official before construction

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

⬡ AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This marine climate demands a well-sealed, moderately insulated envelope that can handle coastal winds up to 92 mph—prioritize impact-resistant glazing and strong connections at corners and edges. Orient primary living spaces and operable windows toward south and east exposures to capture winter sun and morning light, while minimizing west-facing glass to reduce afternoon solar gain in summer months when marine layers eventually burn off. The key passive strategy here is thermal mass paired with natural ventilation: concrete or masonry interior walls will moderate the mild temperature swings, and operable windows on opposite facades will let you flush warm air during the frequent cool coastal breezes rather than relying on mechanical cooling.

⬡ AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This coastal marine site presents well-drained gravelly loam soils suitable for conventional spread footings, though seismic parameters are currently unavailable and may warrant confirmation with the local building official. Moderate wind exposure and modest snow loads are anticipated, suggesting standard structural detailing, but early geotechnical investigation is suggested to establish site class and confirm foundation design approach.

FOUNDATION

Type	*Conventional spread footing anticipated, subject to geotechnical confirmation*
Basis	*Seismic design category currently null - consult local building official and geotechnical engineer to establish applicable SDC and confirm foundation strategy. Well-drained gravelly loam soils suggest spread footing may be viable, pending soil bearing capacity and settlement analysis.*

STRUCTURAL IMPLICATIONS

Seismic	*Seismic parameters unavailable for this coordinate set - consult USGS seismic design maps and local building official to determine SDC, which will drive detailing requirements and structural system selection.*
Wind Exposure	*Exposure C (open terrain with scattered obstructions) anticipated per ASCE 7-22 Risk Category II - verify site surroundings and exposure classification with engineer.*
Wind	*92 mph 3-second gust wind speed anticipated - standard wind load design likely adequate for Risk Category II; no special hurricane provisions anticipated for this inland coastal location.*
Snow	*Ground snow load of 5.8 psf anticipated - standard roof design and drainage likely adequate; consult engineer on roof slope and eave protection given marine winter wind exposure.*

INVESTIGATIONS & INSPECTIONS

Geotech Required	*Yes - geotechnical investigation recommended to confirm soil bearing capacity, settlement characteristics, and to establish site class for seismic design purposes.*
Liquefaction Screen	*Screening recommended - Site Class D soils and marine setting may warrant liquefaction evaluation; consult geotechnical engineer to assess groundwater depth and soil composition relative to liquefaction potential.*
Special Inspections	*May be required depending on final SDC determination - consult building official and structural engineer once seismic design category is established.*
Code Edition	*ASCE 7-22 / IBC 2021 (planning reference only)*
Site Class Note	*Site Class D assumed for planning purposes. Early geotechnical investigation recommended to confirm site class, bearing capacity, and subsurface conditions; establish seismic parameters with local authority.*

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes

SERIES: Coombs

DRAINAGE CLASS: Well drained

HYDRO GROUP: C

SOIL ORDER: Alfisols

SUBORDER: Xeralfs

⬡ AI SOIL INTELLIGENCE — CLAUDE HAIKU

The Coombs gravelly loam is a well-drained Alfisol (Xeralf) formed in alluvial and colluvial materials derived from mixed sources on gentle valley slopes and alluvial fans in a Mediterranean climate. This soil features a gravelly loam surface layer overlying a clay loam or clay substratum, which provides moderate water-holding capacity but can become slowly permeable when saturated. A surveyor or developer should expect moderate excavation difficulty due to gravel content, stable bearing capacity suitable for light structures, and minimal drainage concerns on these 2–5% slopes, though subsoil clay will require proper moisture management if used as fill.

Alluvial and colluvial parent material on valley footslopes and fans; typically vegetated with oak woodland and grassland in semi-arid Mediterranean environment.

DRAINAGE

Class	*Well drained*
Seasonal Saturation	*None within 60 inches*
Flooding Risk	*Minimal on 2–5% slopes; low risk unless adjacent to active channels*
Ponding Risk	*None*

ENGINEERING

Bearing Capacity	*2,500–3,500 psf (shallow foundations)*
Rating	*Good*
Shrink-Swell	*Moderate — clay loam substratum exhibits moderate activity*
Frost Action	*Low*
Cut Suitability	*Good — gravelly loam is stable and workable; remove any organic topsoil first*
Fill Suitability	*Fair — clay loam substratum suitable as fill if moisture conditioned and compacted properly*

SEPTIC

Suitability	*Conditional*
Notes	*Perc test required — subsoil clay loam layer may restrict percolation rates; consider mounding or alternative systems if permeability is insufficient*

HAZARDS

Excavation	*Moderate — 25–35% gravel content may slow digging and require heavy equipment*
Expansive Clay	*Low — clay fraction present but not expansive*
Erosion	*Low on intact slopes; moderate gully and rill erosion risk if grading disturbs surface and drainage is concentrated*

Other	*None anticipated*

Data sources: USGS National Water Information System · waterdata.usgs.gov · California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdmdataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision
Water levels current as of report run: Jul 12, 2026, 8:52 PM UTC
* Live pool elevation not reported by this reservoir — full pool design elevation shown

🛢 OIL & GAS WELLS California — within 10mi radius

8 total | **0** active | **7** plugged

By status: **7** Plugged **1** Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	**Plugged**	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	**Idle**	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	**Plugged**	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil Corporation dry hole located 7.2 miles away, well beyond any actionable distance threshold. No active producers, injection wells, or other well types requiring review are present within relevant distances of the property.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6 Endangered | **6** Threatened | **12** Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	*SCIENTIFIC NAME*	STATUS
California tiger Salamander	*Ambystoma californiense*	**Endangered**
Conservancy fairy shrimp	*Branchinecta conservatio*	**Endangered**
Contra Costa goldfields	*Lasthenia conjugens*	**Endangered**
Soft bird's-beak	*Cordylanthus mollis ssp. mollis*	**Endangered**
Suisun thistle	*Cirsium hydrophilum var. hydrophilum*	**Endangered**
Vernal pool tadpole shrimp	*Lepidurus packardi*	**Endangered**
Alameda whipsnake (=striped racer)	*Masticophis lateralis euryxanthus*	**Threatened**
California red-legged frog	*Rana draytonii*	**Threatened**
Delta smelt	*Hypomesus transpacificus*	**Threatened**
Northern spotted owl	*Strix occidentalis caurina*	**Threatened**
Vernal pool fairy shrimp	*Branchinecta lynchi*	**Threatened**
western snowy plover	*Charadrius nivosus nivosus*	**Threatened**

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Review – 38.3459, -122.3047 **ESA Section 7 Consultation Triggers:** This site's location within critical habitat for 12 listed species triggers mandatory Section 7 consultation if the proposed project involves federal permits, funding, or agency authorization. The presence of six endangered species and six threatened species—particularly vernal pool endemics (California tiger salamander, conservancy fairy shrimp, vernal pool tadpole shrimp) and aquatic species (Delta smelt, California red-legged frog)—indicates high biological sensitivity requiring careful project design review. **Most Planning-Relevant Species:** The vernal pool complex and adjacent wetland species (California tiger salamander, conservancy fairy shrimp, vernal pool tadpole shrimp, and California red-legged frog) present the greatest planning constraints, as these species require intact hydrology and seasonal inundation patterns. The Alameda whipsnake habitat associations should inform upland buffer requirements and vegetation management protocols. **Recommended Next Step:** Commission a Phase I biological assessment and wetland delineation to map critical habitat occurrence and assess project footprint overlap; this assessment should guide site layout, avoidance measures, and potential mitigation strategies before design finalization. Coordinate early with USFWS and CDFW to clarify Section 7 consultation requirements and any project-specific incidental take authorization needs. --- *For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.*

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

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NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57
Unique Species

5051
Total Observations

2026-07-04
Most Recent

COMMON NAME	*SCIENTIFIC NAME*	LAST OBS
American black nightshade	*Solanum americanum*	2026-05-29
Bearded Clover	*Trifolium barbigerum*	2026-05-15
bird's foot cliffbrake	*Pellaea mucronata*	2026-04-28
Blow Wives	*Achyrachaena mollis*	2026-05-20
blue oak	*Quercus douglasii*	2026-05-09
Bolander's Phacelia	*Phacelia bolanderi*	2026-04-27
box elder	*Acer negundo*	2026-05-18
Braun's Giant Horsetail	*Equisetum telmateia braunii*	2026-06-22
California bay	*Umbellularia californica*	2026-06-24
California beeplant	*Scrophularia californica*	2026-05-20
California black oak	*Quercus kelloggii*	2026-05-15
California buckeye	*Aesculus californica*	2026-06-02
California Dutchman's Pipe	*Aristolochia californica*	2026-05-26
California milkwort	*Rhinotropis californica*	2026-05-15
California mugwort	*Artemisia douglasiana*	2026-06-02
California poppy	*Eschscholzia californica*	2026-06-24
California Wild Rose	*Rosa californica*	2026-05-09
Canyon liveforever	*Dudleya cymosa*	2026-05-30
-	*Castilleja densiflora densiflora*	2026-05-09
clay mariposa lily	*Calochortus argillosus*	2026-05-15
coast live oak	*Quercus agrifolia*	2026-05-13
coastal woodfern	*Dryopteris arguta*	2026-05-20
common selfheal	*Prunella vulgaris*	2026-06-22
Diogenes' lantern	*Calochortus amabilis*	2026-05-15
distant phacelia	*Phacelia distans*	2026-05-15
-	*Downingia concolor concolor*	2026-05-15
Dwarf Brodiaea	*Brodiaea terrestris*	2026-05-15
Elegant Clarkia	*Clarkia unguiculata*	2026-05-16
goldback fern	*Pentagramma triangularis*	2026-05-20
Ithuriel's Spear	*Triteleia laxa*	2026-06-15
maroonspot calicoflower	*Downingia concolor*	2026-05-15
narrowleaf milkweed	*Asclepias fascicularis*	2026-06-25
narrowleaf mule-ears	*Wyethia angustifolia*	2026-05-15
Narrowleaf Willow	*Salix exigua*	2026-06-24
needleleaf navarretia	*Navarretia intertexta*	2026-05-15
northern California black walnut	*Juglans hindsii*	2026-06-24
orange bush monkeyflower	*Diplacus aurantiacus*	2026-05-29
Oregon Ash	*Fraxinus latifolia*	2026-06-02
Pacific Bleeding Heart	*Dicentra formosa*	2026-05-04
Pacific madrone	*Arbutus menziesii*	2026-07-04
Pacific poison oak	*Toxicodendron diversilobum*	2026-06-02
pineapple-weed	*Matricaria discoidea*	2026-05-20
Redwood Sorrel	*Oxalis smalliana*	2026-05-28
small-headed clover	*Trifolium microcephalum*	2026-05-15

COMMON NAME	*SCIENTIFIC NAME*	LAST OBS
Superb Mariposa Lily	*Calochortus superbus*	2026-06-11
thimble clover	*Trifolium microdon*	2026-05-15
tomcat clover	*Trifolium willdenovii*	2026-05-15
Toyon	*Heteromeles arbutifolia*	2026-06-19
trailing blackberry	*Rubus ursinus*	2026-06-24
Tree Clover	*Trifolium ciliolatum*	2026-05-15
valley oak	*Quercus lobata*	2026-06-24
wavy-leafed soap plant	*Chlorogalum pomeridianum*	2026-05-20
western sycamore	*Platanus racemosa*	2026-05-17
white brodiaea	*Triteleia hyacinthina*	2026-05-15
white-tipped clover	*Trifolium variegatum*	2026-05-15
Winecup Clarkia	*Clarkia purpurea*	2026-05-15
Yellow Mariposa Lily	*Calochortus luteus*	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment Summary This site in Sonoma County supports a **mixed oak woodland and chaparral community**, dominated by blue oak and California black oak with understory shrubs like California buckeye and California bay. The presence of moisture-loving species (Braun's Giant Horsetail, California Wild Rose) alongside drought-adapted plants (Canyon liveforever, Dudleya) indicates a **transitional mesic-xeric habitat with seasonal water availability**, likely fed by seepage or spring influence. Soils appear to be **moderately well-drained upland areas with intermittent moist microsites**—grading should preserve drainage patterns and avoid compaction in low-lying areas where hydrophytic species concentrate, as disruption could degrade habitat for specialized plants like clay mariposa lily and bird's foot cliffbrake. The high species diversity (57 natives in 10 miles) reflects the ecological sensitivity of this landscape. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

635	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants

VENOMOUS SNAKES

SPECIES	OBSERVATIONS
Crotalus oreganus oreganus	635 obs

VENOMOUS SPIDERS

SPECIES	OBSERVATIONS
Latrodectus hesperus	154 obs

STINGING INSECTS

SPECIES	OBSERVATIONS
Dasymutilla aureola	91 obs
Dasymutilla californica	91 obs
Dasymutilla sackenii	91 obs

TOXIC PLANTS

SPECIES	OBSERVATIONS
Toxicodendron diversilobum	1100 obs
Datura wrightii	57 obs
Datura stramonium	57 obs
Datura ferox	57 obs
Conium maculatum	199 obs
Urtica gracilis	83 obs
Urtica gracilis holosericea	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Briefing – 38.3459, -122.3047 Your highest-priority hazards are poison oak (Toxicodendron diversilobum)—extremely common in this region with 1,439 observations—and Northern Pacific rattlesnakes (Crotalus oreganus oreganus) with 635 documented sightings. Wear long pants, closed-toe boots, and heavy gloves; wash exposed skin immediately after fieldwork and launder all clothing separately to avoid oil transfer. Stay alert for rattlesnakes in tall grass and rocky areas, especially during warm months (April–October), and check equipment and gear before packing. Secondary concerns include widow spiders and toxic daturas in disturbed areas, so avoid touching unfamiliar plants and shake out equipment before use. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3 Beam Paths | **545ft** Closest Beam | **3** Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26 Total Airways | **13** Jet Routes | **2** Victor Airways | **11** Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

8 Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200	91	66	5	14	6
Total Observations	Total Species	Birds	Mammals	Reptiles	Amphibians

BIRDS (66 species)

COMMON NAME	*Scientific Name*	OBS
Wild Turkey	*Meleagris gallopavo*	9
Snowy Egret	*Egretta thula*	7
California Quail	*Callipepla californica*	5
Western Bluebird	*Sialia mexicana*	5
California Towhee	*Melozone crissalis*	5
Great Egret	*Ardea alba*	4
Red-tailed Hawk	*Buteo jamaicensis*	4
Black Phoebe	*Sayornis nigricans*	4
Western Screech-Owl	*Megascops kennicottii*	4
Cooper's Hawk	*Astur cooperii*	3
Anna's Hummingbird	*Calypte anna*	3
Ash-throated Flycatcher	*Myiarchus cinerascens*	3
Turkey Vulture	*Cathartes aura*	3
Mallard	*Anas platyrhynchos*	3
Bushtit	*Psaltriparus minimus*	3
+ 51 additional species		

MAMMALS (5 species)

COMMON NAME	*Scientific Name*	OBS
Black-tailed Jackrabbit	*Lepus californicus*	4
Coyote	*Canis latrans*	2
Mule Deer	*Odocoileus hemionus*	2
Eastern Fox Squirrel	*Sciurus niger*	1
Striped Skunk	*Mephitis mephitis*	1

REPTILES (14 species)

COMMON NAME	*Scientific Name*	OBS
Western Fence Lizard	*Sceloporus occidentalis*	9
Pacific Gopher Snake	*Pituophis catenifer catenifer*	8
Northern Pacific Rattlesnake	*Crotalus oreganus oreganus*	6
California King Snake	*Lampropeltis californiae*	3
Pond Slider	*Trachemys scripta*	2
Western Skink	*Plestiodon skiltonianus*	2
Ring-necked Snake	*Diadophis punctatus*	1
Common Garter Snake	*Thamnophis sirtalis*	1
Western Yellow-bellied Racer	*Coluber constrictor mormon*	1
Pacific Ringneck Snake	*Diadophis punctatus amabilis*	1
Northwestern Pond Turtle	*Actinemys marmorata*	1
Common Sharp-tailed Snake	*Contia tenuis*	1
North American Racer	*Coluber constrictor*	1
Southern Alligator Lizard	*Elgaria multicarinata*	1

AMPHIBIANS (6 species)

COMMON NAME	*Scientific Name*	OBS
Pacific chorus frog	*Pseudacris regilla*	5
American Bullfrog	*Lithobates catesbeianus*	3
Foothill Yellow-legged Frog	*Rana boylii*	3
Rough-skinned Newt	*Taricha granulosa*	1
Western Toad	*Anaxyrus boreas*	1
California Giant Salamander	*Dicamptodon ensatus*	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment – 38.3459, -122.3047 This site supports a moderately diverse wildlife community indicative of oak woodland and grassland habitat with proximal freshwater resources, likely within the North Bay region of California. The presence of sensitive species including Foothill Yellow-legged Frog and California Giant Salamander suggests seasonal wetlands or reliable riparian corridors requiring hydrological sensitivity, while the diverse reptile assemblage and breeding songbirds (Western Bluebird, California Towhee) indicate relatively intact scrub-grassland conditions with mature structural elements. The mammalian fauna and raptor diversity (Red-tailed Hawk, Cooper's Hawk, Western Screech-Owl) reflect a landscape mosaic with moderate connectivity; however, the presence of non-native species (Eastern Fox Squirrel, American Bullfrog, Pond Slider) suggests localized anthropogenic disturbance. Overall, this community characterizes a semi-rural transitional zone with intact but fragmented habitat connectivity—development feasibility should be evaluated with particular attention to aquatic and amphibian habitat protection, oak woodland retention, and maintenance of wildlife corridor function. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

☀ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = **38.3°** (latitude) → **9/12 pitch**

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	← excellent
7/12	30.3°	98.8%	← excellent
8/12	33.7°	99.3%	← excellent
9/12	36.9°	99.8%	← annual optimal
10/12	39.8°	99.8%	← excellent
12/12	45°	99%	← excellent

Going from 5/12 to 6/12 at framing costs ~$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~$108 value/yr). Over 25yr panel life: ~$2710 total gain.

3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	⚠ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	⚠ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr·ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 · ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 · THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 · TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK

20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	⚠ verify
50ft	All year	4.8	4.6	7.3	⚠ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.
⚠ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 · ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45
Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45
Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor
†Balanced = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)
WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.
⚠ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

⬡ AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan Your location gets about 6.6 hours of useful sunlight per day on average—think of it like having 6.6 hours of "charging time" for solar panels each day, which is solid for the California coast, though it dips to just 3.2 hours in your darkest month. Your roof's pitch is nearly perfect for capturing that sun (it's already angled at about 38 degrees, which is ideal), so a solar installation here would work really well. Now, here's something most homeowners overlook: a dark roof acts like an oven on your house in summer, pulling in about 300 units of heat per hour from every square foot of roofing—but if you switched to a light-colored roof, that would drop to about 110, which is huge for cooling costs and comfort. Your west-facing windows are the real troublemaker in afternoon heat, letting in roughly twice the summer heat load of south-facing glass, so shading those (or upgrading the glass) should be a priority. A big deciduous tree planted on the south side is your friend here—it'll block the sun for about 4 hours during summer when you need it, but lose its leaves and let warming sun through during the 5 winter hours when that free heat helps you out. Think of thermal mass like a sponge: it's the heavy materials in your home (concrete, brick, thick walls) that slowly absorb heat during the day and release it at night, smoothing out temperature swings—the better your walls, the longer that happens. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.

🧠 **SITE INTELLIGENCE SUMMARY** AI synthesis — Claude Sonnet

⚙ Technical 👤 Client

Technical Brief — for architects, engineers, and surveyors

SITE INTELLIGENCE BRIEFING **38.34586°N, 122.30466°W | Napa County, California**

Structural Design Drivers

This site presents an atypical structural control regime for northern California. Wind and snow loading are not quantified in the available database, and the IECC Zone 3C (Marine) climate classification with 0°F-days Annual Freezing Index indicates minimal frost-related loading. Seismic parameters are not reported in this dataset, which is a critical gap for Napa County given its known tectonic activity. Based on the data provided, live load and thermal cycling emerge as the primary documented design drivers, though the absence of seismic coefficients and wind speed data suggests that further investigation into CBC Section 11 (Seismic) and Section 12 (Interior Environment) requirements is mandatory before finalizing the lateral system.

Geotechnical Foundation Strategy

The site soil is mapped as Coombs gravelly loam, 2 to 5 percent slopes, classified as Hydrologic Group C (moderate runoff potential) with well-drained characteristics. This profile supports shallow foundation design; the gravelly loam matrix and good drainage indicate low settlement risk and no perched water table concerns. The reported frost depth is not available, but the 0°F-days AFI signals minimal heave risk—footings can be set shallower than in continental climates, typically 12–18 inches below finished grade in Napa County practice. A site-specific geotechnical investigation must confirm bearing capacity (estimated 2,500–3,500 psf for gravelly loam, pending lab confirmation), confirm the absence of subsurface clay lenses, and validate drainage performance on a 2–5 percent slope.

Hazard Assessment

No Quaternary faults are recorded within the search radius, and no subsidence basin data applies to this location. The eight oil and gas wells within approximately 10 miles are all plugged or non-producing (nearest Mobil well 7.2 miles distant), eliminating subsidence and induced seismicity concerns from active extraction. However, the absence of seismic hazard data in this briefing does not negate seismic risk; the design team must obtain CalGEO Earthquake Hazards Zone mapping and fault rupture/strong-motion parameters separately. Wildfire hazard data was not provided; verify California State Responsibility Area classification and defensible space requirements if applicable.

Site Context and Climate Envelope

The IECC Zone 3C (Marine) classification denotes a mild, coastal-influenced climate with minimal heating and cooling demand—typical of sheltered valleys in Napa. The 0°F-days AFI confirms no design winter ground temperature penalty. The 2–5 percent topographic slope supports good surface drainage and reduces ponding risk on grade. No mineral deposits or active mining operations are present within the search radius. The proximity to eight nonproducing oil and gas wells should be flagged for title review and Phase I ESA cross-reference, though current hazard is low.

Bottom Line

Initiate immediate geotechnical site investigation (soil borings, laboratory testing, bearing capacity analysis) and obtain seismic hazard parameters and current CBC wind/seismic coefficients from the local building department before finalizing foundation design or lateral-system strategy.

📋 Copy to clipboard

Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

SITE SUMMARY

Coordinates	38.345860°N 122.304660°W
Elevation	66 ft NAVD88 (approx)
Township/Range	T6N, R4W, Mount Diablo Meridian
FEMA Zone	Zone X Panel 06055C0510FF
Nearest BM	JT9010 0.2mi 62.7ftft
Nearest CORS	P261 14.1mi
State Plane	CA 2 (0402)

TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

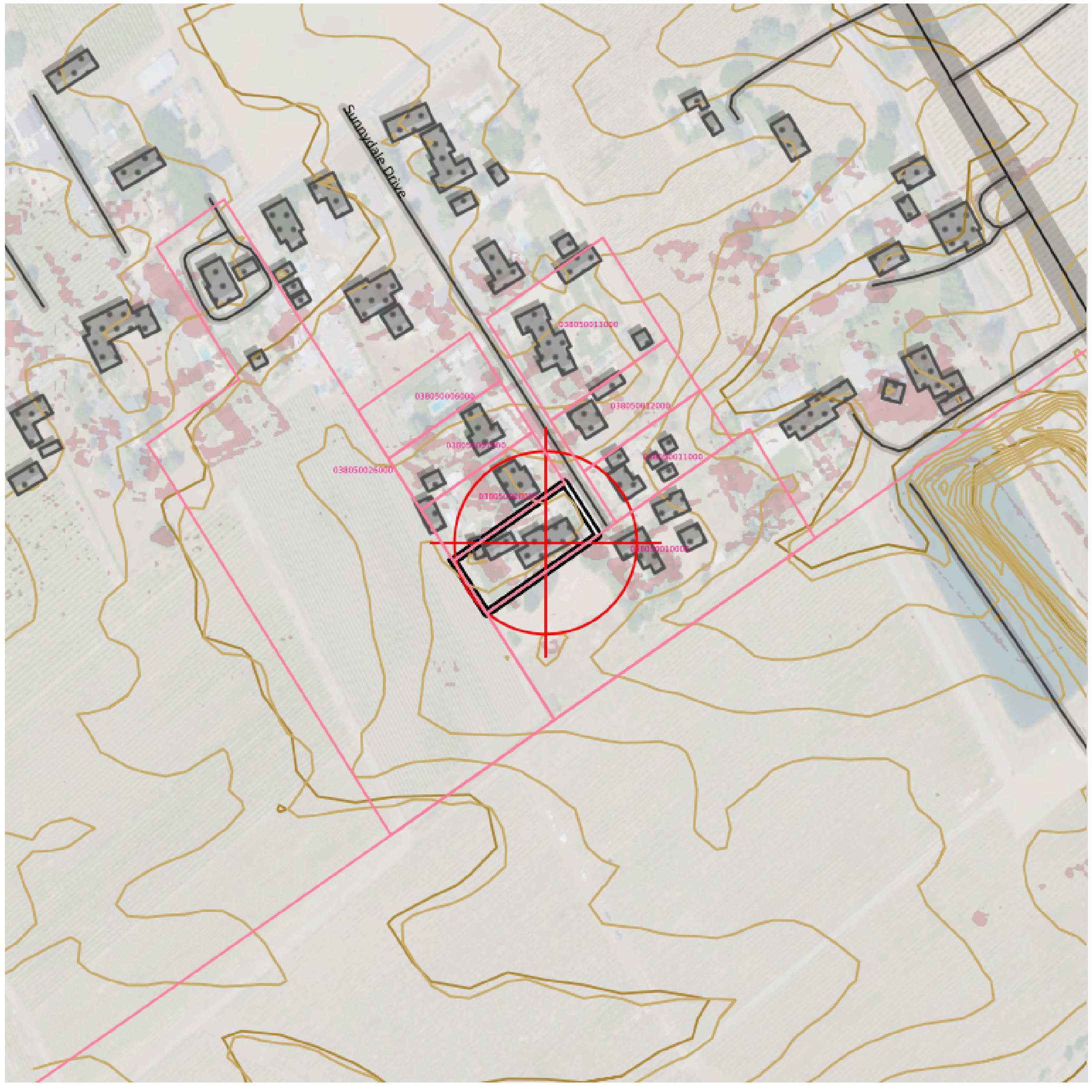
Generated: July 12, 2026
Base: USGS National Map
Generated by TopoScout · outKastdesigns.com

TOPOSCOUT SITE DXF PREVIEW

45 CAD Layers · AutoCAD Compatible · State Plane Coordinates

$19 UPGRADE → GET THE .DXF FILE

TOPOSCOUT SITE PLAN PREVIEW | $19 UPGRADE FOR AUTOCAD DXF FILE



INCLUDED LAYERS

N

0 —— 400 ft State Plane · NAD83 · US Feet

■ Contours (Index/Major/Minor/1ft) ■ PLSS Section/Township/QQ ■ Roads Major/Local/Trail + ROW ■ Buildings ■ Hydro ■ Power Lines ■ Pipeline ■ Railroad ■ Benchmarks + Labels
■ County Boundary ■ OTLS Grid ■ Site Info ■ Elevation Labels ■ 3D Polylines (all geometry)

⚡ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online – https://trimblegnssplanningonline.com/
Satellite visibility, DOP charts, sky plots

FAA B4UFLY – https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly
Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – https://www.aloft.ai/
Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp – https://www.swpc.noaa.gov/products/planetary-k-index
Real-time Kp index, geomagnetic storm alerts

BASE STATION & NETWORK

NOAA CORS Network – https://geodesy.noaa.gov/CORS/cors-data.shtml
Continuously operating reference stations, data download

NGS CORS Station Map – https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml
Interactive map of all CORS stations with status

IGS Station Monitor – https://www.igs.org/network
International GNSS Service — global base station health

POST-PROCESSING

NGS OPUS – https://geodesy.noaa.gov/OPUS/index.jsp
Online Positioning User Service — static GNSS processing

NGS OPUS-RS – https://geodesy.noaa.gov/OPUSI/index.jsp
Rapid static processing — sessions as short as 15 minutes

CORS Data Download – https://geodesy.noaa.gov/CORS/cors-data.shtml
Raw RINEX observation files from CORS stations

REFERENCE & CALCULATIONS

NGS Datasheets – https://www.ngs.noaa.gov/cgi-bin/datasheet.prl
Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) – https://geodesy.noaa.gov/NCAT/
NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination – https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml
Declination calculator for any location and date

NGS Geoid Models – https://geodesy.noaa.gov/GEOID/
GEOID18 and current model downloads, GEOID calculator

FEMA & FLOOD

FEMA Flood Map Service Center – https://msc.fema.gov/portal/home
FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate
Current EC form, instructions, LOMA/LOMR submission guidance

DRONE OPERATIONS

FAA Part 107 — Commercial Rules – https://www.faa.gov/uas/commercial_operators
Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – https://faadronezone.faa.gov/
Drone registration, operator certificates, waiver management

FAA B4UFLY – https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly
Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers – https://www.faa.gov/uas/commercial_operators/part_107_waivers
Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – https://www.aloft.ai/
Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

AirNow Fire & Smoke Map
https://www.fire.airnow.gov
Current fire locations, smoke plumes, PM2.5 readings by location

AirNow Interactive Map
https://gispub.epa.gov/airnow
Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

NOAA Smoke Forecast
https://airquality.weather.gov
48-hr smoke concentration forecast, HRRR-Smoke model